

Test Plan Document

HOSTEL MANAGEMENT SYSTEM

Project Code: _____

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Submission Date: _____

Project Manager's Signature

Document Information

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Definition of Terms, Acronyms and Abbreviations

Term	Description
HMS	Hostel Management System
SRS	Software Requirements Specifications
UAT	User Acceptance Testing

1. Introduction

The Hostel Management System is developed to manage hostel operations such as student registration, room allocation, fee management, mess services, and complaint handling. This Test Plan document describes the approach, scope, and methods used to test the system in order to ensure that it works correctly and meets the specified requirements.

1.1 Purpose of Document

- The purpose of this Test Plan document is to describe the testing approach for the Hostel Management System.
- It explains how the system will be tested to ensure that all functions work correctly and meet the requirements defined in the Software Requirements Specification (SRS).
- This document is intended for project team members, testers, and supervisors who are involved in testing and evaluating the system.

1.2 Project Overview

- The Hostel Management System is a software application designed to manage hostel-related activities such as student registration, room allocation, fee management, mess management, and complaint handling.
- The system is used by different users including students, warden, accountant, mess in-charge, and administrator.
- Testing of this system focuses on verifying that each module performs its intended functions accurately and that the system works smoothly as a whole.

2. Scope of Testing

This section defines what parts of the Hostel Management System will be tested and what will not be included in testing.

2.1 Features to be Tested

The following features of the Hostel Management System will be included in testing:

- **Registration Module**
 - Student account creation
 - Login and authentication
 - Updating student profile information
- **Student Module**
 - Hostel admission form submission

- Room request and allocation
- Fee and payment records
- Mess menu enrollment
- Complaint submission and tracking
- **Warden Module**
 - Review and approval/rejection of student applications
 - Room allocation and management
 - Student attendance marking
 - Complaint handling
 - Notice and rule updates
- **Room Module**
 - Room availability and status
 - Room type and capacity management
- **Accountant Module**
 - Fee record management
 - Fee report generation
 - Updating hostel fee charges
- **Mess In-charge Module**
 - Daily mess menu management
 - Mess attendance records
 - Mess-related reports
- **Admin Module**
 - User account management
 - Hostel data management
 - System backup and restore
 - Activity monitoring

2.2 Features Not to be Tested

The following features are not included in the current testing scope:

- Hardware or network performance testing
- Mobile application testing (as system is web-based only)

3. Test Plan Strategy

Testing of the Hostel Management System will be carried out using a structured approach. Each module will be tested individually to verify its functionality, followed by integration testing to ensure that all modules work together properly. Both positive and negative test cases will be used to validate system behavior. Manual testing will be primarily used to check correctness, usability, and data validation.

3.1 Unit Testing

Individual modules such as Student, Warden, Accountant, and Mess modules will be tested separately.

3.2 Integration Testing

Interaction between modules (e.g., Student and Room module, Warden and Fee module) will be tested.

3.3 System Testing

The complete system will be tested as a whole to ensure it meets the requirements.

3.4 User Acceptance Testing (UAT)

The system will be tested from the user's perspective to confirm it is ready for use.

4. Test Environment

Testing of the HMS will be performed in a controlled environment that is similar to the actual usage setup. The system will be tested on standard computers with the required software installed. A test database will be used to avoid affecting real data. Testing will be carried out using common web browsers and basic system configurations.

4.1 Hardware Requirements

- Personal computer or laptop
- Minimum 4 GB RAM
- Stable internet connection

4.2 Software Requirements

- Operating System: Windows
- Web Browser: Google Chrome / Microsoft Edge
- Database: MySQL
- Development Environment: Local server or IDE used for development

5. Schedule

Testing Activities	Begin	End
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Test Case Design	20-01-2026	22-02-2026
Unit Testing	23-01-2026	25-01-2026
Integration Testing	dd-mm-yyyy	dd-mm-yyyy
System Testing	dd-mm-yyyy	dd-mm-yyyy
User Acceptance Testing	dd-mm-yyyy	dd-mm-yyyy
Test Report Preparation	dd-mm-yyyy	dd-mm-yyyy

6. Control Activities

- Regular review meetings will be conducted with the project team and supervisor.
- The test team will report progress and issues weekly.
- Any critical bugs will be immediately discussed and fixed before moving to the next phase.

7. Functions to be Tested

The following functions will be tested:

- Student registration and login
- Student profile updates
- Hostel admission form submission
- Room request and allocation
- Fee calculation, payment, and reports
- Mess menu and attendance management
- Complaint submission and tracking
- Warden approval of requests and attendance marking
- Admin account and hostel data management
- Backup and restore functions

8. Functions not to be Tested

- Hardware performance (CPU/RAM stress testing)
- Mobile application features
- Third-party email or notification service reliability

9. Test Case Design and Description

Each test case includes:

Attribute	Description
Test Case ID	Unique identifier
Objective	What is being tested

Module	Which HMS module
Inputs	Data entered during test
Expected Output	Result expected
Actual Output	Result during test
Status	Pass / Fail / Not Executed

Test Case Template

Module Name: Login

Test Case ID: hmslogin-01

QA Test Engineer: Alishba, Arsheen, Samiya

Test case Version:01

Test Date: 23-01-2026

Objective: To verify that only registered users with correct credentials can access the system and that unauthorized users are properly blocked.

Environment: Windows OS, web browser (Chrome/Microsoft Edge), MySQL database, stable internet connection

Assumptions: User has valid credentials, database is running, system is accessible

Pre-Requisite: User account must be registered in the system before attempting login

Result: Pass

10. Traceability Matrix

The traceability matrix shows the relationship between requirements, use cases, and test cases to ensure forward and backward traceability.

Sr. no.	Requirement ID	Use Case ID	Test case ID	Status	Date Completed
1	REQ-001	UC-001	TC-001	Complete	23-01-2026
2	REQ-002	UC-002	TC-002	Incomplete	dd-mm-yyyy

11. Major Deliverables

The following items will be delivered at the end of the test phase:

- **Test Plan Document:** This document describing the overall testing strategy and approach

- **Test Cases:** Detailed test cases for all modules and functions
- **Test Execution Reports:** Results of test case execution showing pass/fail status
- **Bug Reports:** Documentation of all bugs found during testing
- **Test Summary Report:** Final report summarizing testing activities, results, and recommendations

12. Risks and Assumptions

Assumptions

The following assumptions have been taken into consideration when developing this test schedule:

- Unit testing will be completed prior to receiving code for system testing
- All testing personnel have a good knowledge of the system
- Computers and required software will be available during all working hours
- Test data will be available when needed

Risks

The following risks may affect the testing process:

- Delays in development may reduce time available for testing
- Changes in requirements during testing phase
- Limited availability of key personnel during testing
- Technical issues with test environment setup
- Incomplete or unclear requirements documentation

13. Exit Criteria

Testing will be considered complete when the following criteria are met:

- All functions identified in the requirement document are present and working
- All planned test cases have been executed
- Test summary report has been prepared and approved
- All identified critical bugs have been fixed and retested
- System meets all requirements specified in the SRS document

14. References

This section provides a complete list of all documents referenced during the preparation of this test plan.

Ref. No.	Document Title	Date of Release/Publication	Document Source
PGBH01-2003-SRS	Software Requirements Specifications	Oct 3, 2025	Project repository/folder
PGBH01-2003-Design	Software Design Specification	Oct 3, 2025	Project repository/folder
PGBH01-2003-Proposal	Project Proposal		Project repository/folder

15. Appendices

Sample Test Cases

Test Case ID: TC-001

Module: Registration Module

Objective: Verify that a new student can successfully register

Step No.	Execution Description	Expected Result	Status
1	Open registration page	Registration form is displayed	
2	Enter valid student details	Fields accept input	
3	Click Submit button	Success message displayed	
4	Verify account created	Student can login	

Test Case ID: TC-002

Module: Student Module

Objective: Verify that student can submit hostel admission form

Step No.	Execution Description	Expected Result	Status
1	Login as student	Dashboard is displayed	
2	Navigate to admission form	Form page loads	
3	Fill all required fields	Fields accept input	

4	Submit the form	Confirmation message shown	
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